

PHYS 230 - Electricity and Magnetism

Lecture 10

Instructor: Saeed Rastgoo



**UNIVERSITY
OF ALBERTA**

Outline

2 Gauss's Law

- Gauss's Law (Continued)

3 Electric Potential Energy and Potential

- Reminder: Work and Energy
- Electric Potential Energy

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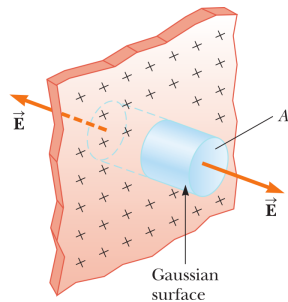
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Exercise 7

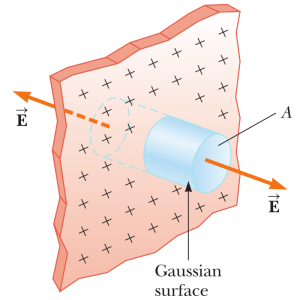
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Find the electric field due to an infinite plane of positive charge with uniform surface charge density σ .



Exercise 7

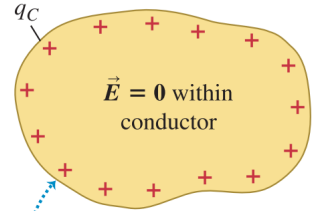
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Charges On a Conductor

We learned that $\mathbf{E} = 0$ inside a conductor $\Rightarrow q_{\text{in}} = 0$

(a) Solid conductor with charge q_C

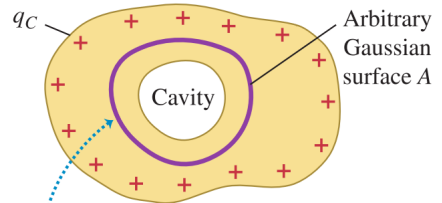


The charge q_C resides entirely on the surface of the conductor. The situation is electrostatic, so $\vec{E} = 0$ within the conductor.

A Conductor With a Cavity

- Let's create a cavity inside the conductor

(b) The same conductor with an internal cavity

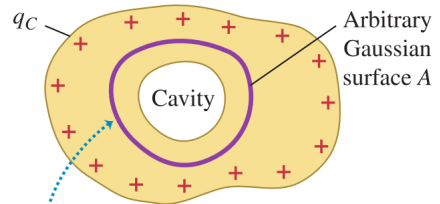


Because $\vec{E} = \mathbf{0}$ at all points within the conductor, the electric field at all points on the Gaussian surface must be zero.

A Conductor With a Cavity

- Let's create a cavity inside the conductor
- Still $\mathbf{E} = 0$ inside a conductor $\Rightarrow q_{\text{in}} = 0 \Rightarrow$ there cannot be any net charges on the surface of cavity

(b) The same conductor with an internal cavity

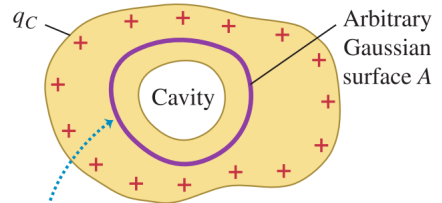


Because $\vec{E} = 0$ at all points within the conductor, the electric field at all points on the Gaussian surface must be zero.

Important Result

- Let's imagine: remove the conductor all together but keep the charges where they are

(b) The same conductor with an internal cavity

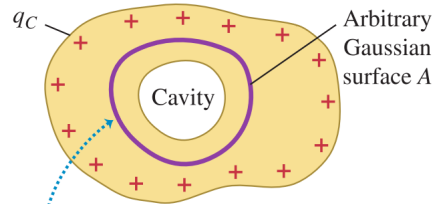


Because $\vec{E} = \mathbf{0}$ at all points within the conductor, the electric field at all points on the Gaussian surface must be zero.

Important Result

- Let's imagine: remove the conductor all together but keep the charges where they are
- Still $\mathbf{E} = 0$ inside the boundary where charges lie

(b) The same conductor with an internal cavity



Because $\vec{E} = 0$ at all points within the conductor, the electric field at all points on the Gaussian surface must be zero.

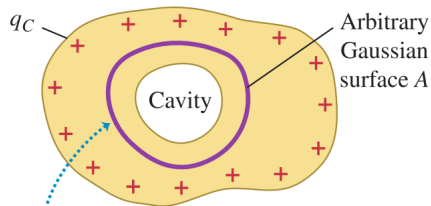
Important Result

- Let's imagine: remove the conductor all together but keep the charges where they are
- Still $\mathbf{E} = 0$ inside the boundary where charges lie

Only the charges inside a closed surface contribute to $\mathbf{E}$ on the surface

Charge outside the surface have no effect whatsoever!

(b) The same conductor with an internal cavity

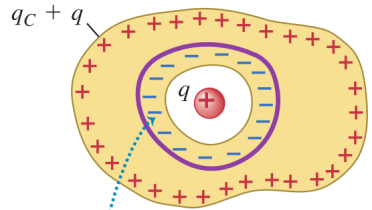


Because $\vec{E} = 0$ at all points within the conductor, the electric field at all points on the Gaussian surface must be zero.

A Conductor With a Cavity And a Charge Within It

- Let's put a charge q inside the cavity

(c) An isolated charge q placed in the cavity

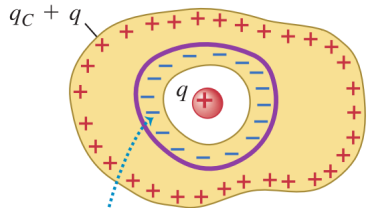


For $\vec{E}$ to be zero at all points on the Gaussian surface, the surface of the cavity must have a total charge $-q$.

A Conductor With a Cavity And a Charge Within It

- Let's put a charge q inside the cavity
- Still $\mathbf{E} = 0$ inside a conductor $\Rightarrow q_{\text{in}} = 0 \Rightarrow$ there would be induced opposite charges $-q$ right on the surface of cavity

(c) An isolated charge q placed in the cavity



For $\vec{E}$ to be zero at all points on the Gaussian surface, the surface of the cavity must have a total charge $-q$.

E At The Surface of a Conductor

- **E** at the surface of a conductor is **always perpendicular to the surface**, otherwise currents would exist on the surface of conductor (due to force $\mathbf{F} = q\mathbf{E}$ on free electrons)

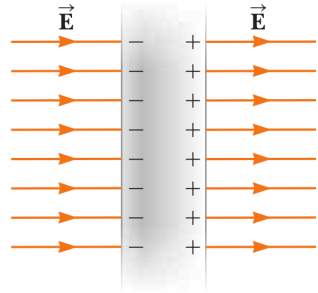


Figure 24.16 A conducting slab in an external electric field $\vec{E}$. The charges induced on the two surfaces of the slab produce an electric field that opposes the external field, giving a resultant field of zero inside the slab.

E At The Surface of a Conductor

- **E** at the surface of a conductor is **always perpendicular to the surface**, otherwise currents would exist on the surface of conductor (due to force $\mathbf{F} = q\mathbf{E}$ on free electrons)
- Magnitude of **E** is

$$E = \frac{\sigma}{\epsilon_0}$$

because by Gauss's law (look at figure)

$$\int_{\text{top}} \mathbf{E}_{\text{top}} \cdot d\mathbf{A}_{\text{top}} + \underbrace{\int_{\text{bottom}} \mathbf{E}_{\text{bottom}} \cdot d\mathbf{A}_{\text{bottom}}}_0 + \underbrace{\int_{\text{side}} E_{\text{side}} dA_{\text{side}} \cos(90)}_0 = \frac{q_{\text{in}}}{\epsilon_0}$$

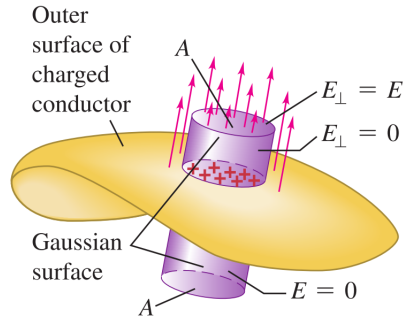
or since $q_{\text{in}} = \sigma A_{\text{top}}$ and $\mathbf{E}_{\text{top}} = E\hat{\mathbf{n}}$ with constant E , and $d\mathbf{A}_{\text{top}} = dA_{\text{top}}\hat{\mathbf{n}}$, we get

$$EA_{\text{top}} = \frac{\sigma A_{\text{top}}}{\epsilon_0}$$

and thus

$$E = \frac{\sigma}{\epsilon_0}$$

Figure 22.28 The field just outside a charged conductor is perpendicular to the surface, and its perpendicular component $E_{\perp}$ is equal to σ/ϵ_0 .



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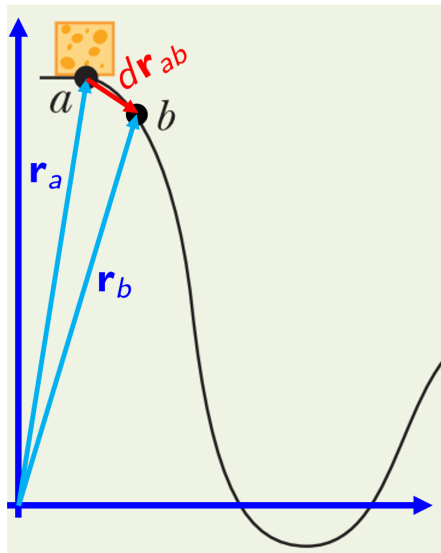
Reminder: Work

The infinitesimal work dW of a force $\mathbf{F}$ on an object going from a point a with position vector $\mathbf{r}_a$ to a point b with position vector $\mathbf{r}_b$ is

$$dW = \mathbf{F} \cdot d\mathbf{r} = F dr \cos(\theta_{\mathbf{F}, d\mathbf{r}})$$

where

$$d\mathbf{r} = \mathbf{r}_b - \mathbf{r}_a$$

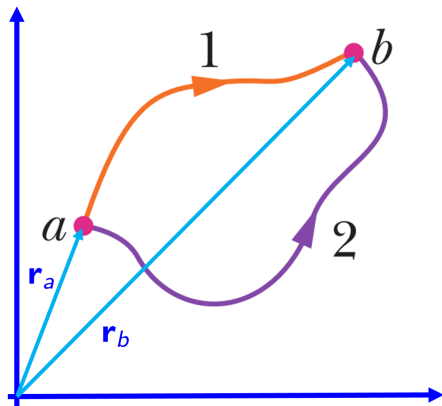


Reminder: Work

The finite work W of a force $\mathbf{F}$ on an object going from a point a with position vector $\mathbf{r}_a$ to a point b with position vector $\mathbf{r}_b$ on a path 1 is

$$W_{ab}^{(1)} = \int_a^b dW = \int_{\mathbf{r}_a}^{\mathbf{r}_b} \mathbf{F} \cdot d\mathbf{r} = \int_{\mathbf{r}_a}^{\mathbf{r}_b} F \cos(\theta_{\mathbf{F}, d\mathbf{r}}) dr$$

and it is understood that the integral is taken over path 1.



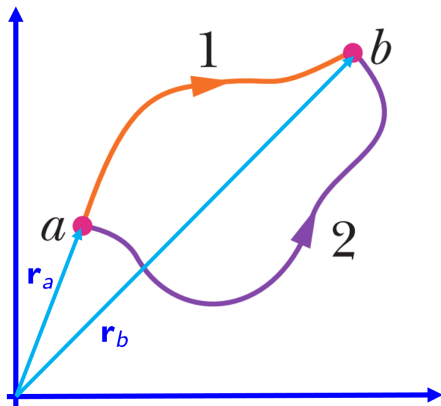
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- This in general depends on the path taken



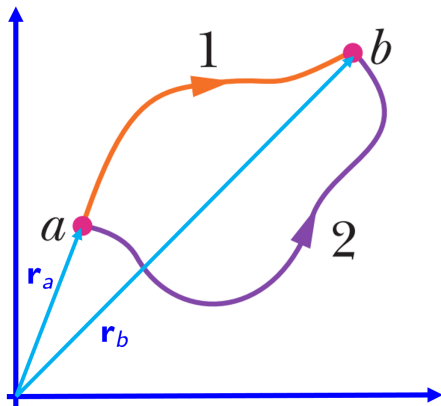
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and it is understood that the integral is taken over path 1.

- This in general depends on the path taken
- So in general $W_{ab}^{(1)} \neq W_{ab}^{(2)}$

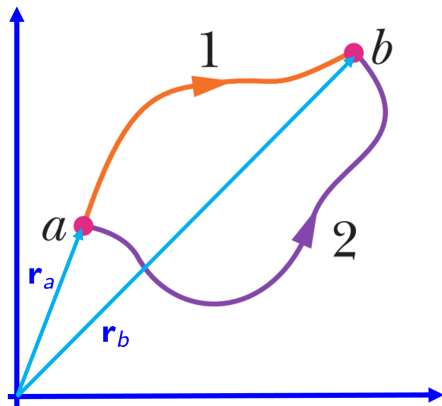


Reminder: Conservative Forces

Conservative Force $\mathbf{F}$: one for which **the work** on an object going from a point a with position vector $\mathbf{r}_a$ to a point b with position vector $\mathbf{r}_b$ **is path-independent**

$$W_{ab}^{(1)} = W_{ab}^{(2)} = \boxed{W_{ab} = \int_{\mathbf{r}_a}^{\mathbf{r}_b} \mathbf{F} \cdot d\mathbf{r}} = \int_{\mathbf{r}_a}^{\mathbf{r}_b} F \cos(\theta_{\mathbf{F}, d\mathbf{r}}) dr$$

where W_{ab} with no superindex means along any path from a to b .



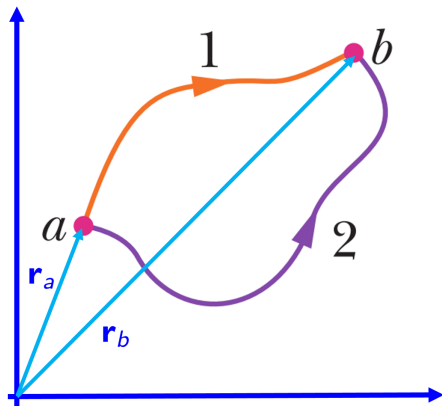
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- For conservative forces, **work only depends on the initial and final points** not on the path taken



Reminder: Conservative Forces

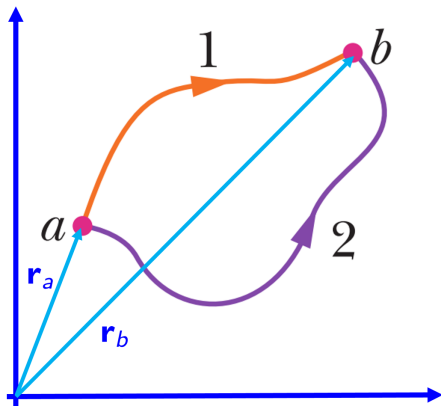
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where W_{ab} with no superindex means along any path from a to b .

- For conservative forces, **work only depends on the initial and final points** not on the path taken
- This also means that the work along any closed path is zero!

$$W_{ab}^{(1)} + W_{ba}^{(2)} = \int_{\mathbf{r}_a}^{\mathbf{r}_b} \mathbf{F} \cdot d\mathbf{r} + \int_{\mathbf{r}_b}^{\mathbf{r}_a} \mathbf{F} \cdot d\mathbf{r} = \int_{\mathbf{r}_a}^{\mathbf{r}_b} \mathbf{F} \cdot d\mathbf{r} - \int_{\mathbf{r}_a}^{\mathbf{r}_b} \mathbf{F} \cdot d\mathbf{r} = 0$$



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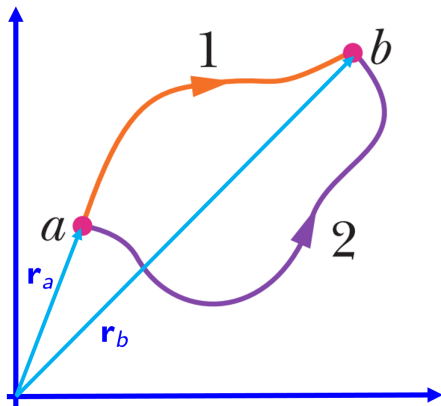
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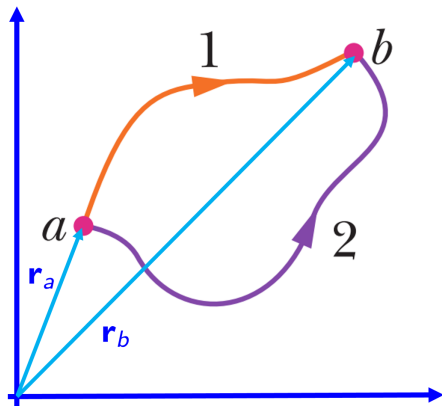
- **Gravity and electric forces are conservative.**



Reminder: Potential Energy

- For any conservative force $\mathbf{F}$ we can define a **potential energy** U such that for any infinitesimal displacement $d\mathbf{r}$ of the object on which the force acts, we have

$$\boxed{dU = -dW} \Rightarrow dU = -\mathbf{F} \cdot d\mathbf{r} = -F dr \cos(\theta_{\mathbf{F}, d\mathbf{r}})$$



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- For a displacement on a finite path from point a to point b we have

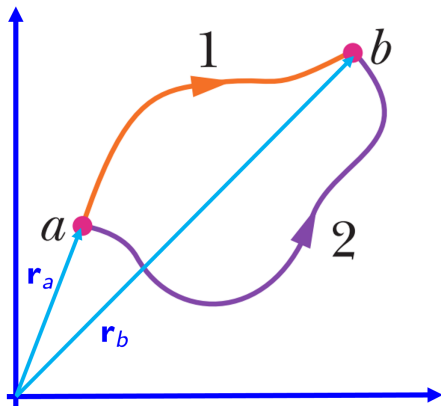
$$\int_{\mathbf{r}_a}^{\mathbf{r}_b} dU = - \int_{\mathbf{r}_a}^{\mathbf{r}_b} dW$$

or

$$\boxed{U_b - U_a = - \int_{\mathbf{r}_a}^{\mathbf{r}_b} \mathbf{F} \cdot d\mathbf{r}}$$

or

$$U_b - U_a = -W_{ab}$$

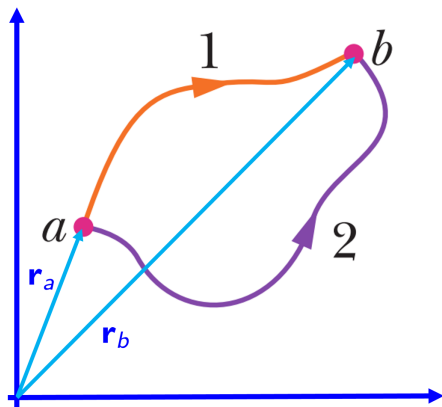


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- Note: Potential energy is a **property of CONFIGURATION of a SYSTEM**, not just one object!

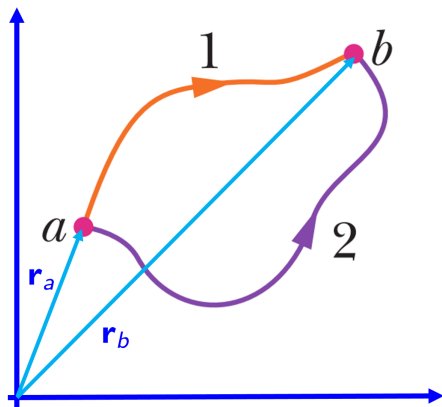


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 - In this case the property of the force field $\mathbf{F}$ (or the object creating the force field) AND the object on which the force is exerted

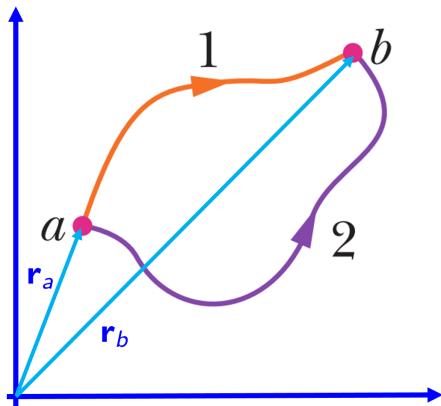


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- Note: Potential energy is a **property of CONFIGURATION of a SYSTEM**, not just one object!
 - In this case the property of the force field $\mathbf{F}$ (or the object creating the force field) AND the object on which the force is exerted
 - Always need at least two objects to have a potential energy NOT ONE



Reminder: Conservation of Energy

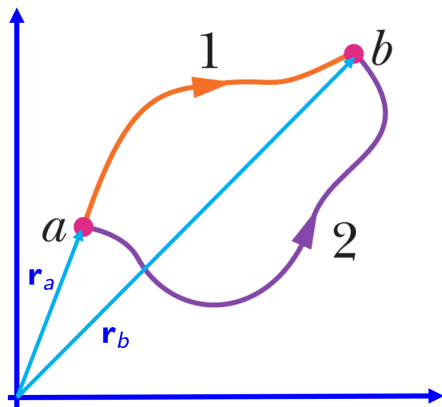
For any system where only conservative forces $\mathbf{F}$ exist, the total mechanical energy $E = K + U$ is conserved

$$K_a + U_a = K_b + U_b$$

where

$$K = \frac{1}{2}mv^2$$

is the kinetic energy of the object at some point (a or b).



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Electric Potential Energy

- Take a charge q in a field $\mathbf{E}$

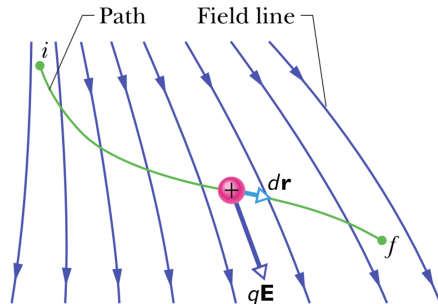


Fig. 24-4 A test charge q moves from point i to point f along the path shown in a nonuniform electric field. During a displacement $d\mathbf{r}$, an electrostatic force $q\mathbf{E}$ acts on the test charge. This force points in the direction of the field line at the location of the test charge.

Electric Potential Energy

- Take a charge q in a field $\mathbf{E}$
- A force $\mathbf{F} = q\mathbf{E}$ (conservative) acts on the object

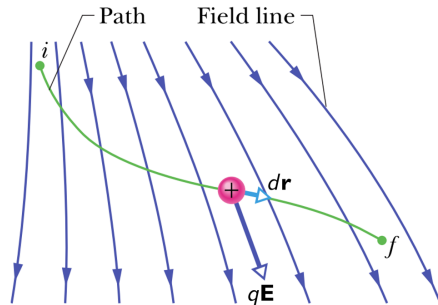


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Electric Potential Energy

- Take a charge q in a field $\mathbf{E}$
- A force $\mathbf{F} = q\mathbf{E}$ (conservative) acts on the object
- If it goes from $\mathbf{r}_a$ to $\mathbf{r}_b$, the **change in its potential energy in general** is

$$U_b - U_a = - \int_{\mathbf{r}_a}^{\mathbf{r}_b} \mathbf{F} \cdot d\mathbf{r} \Rightarrow U_b - U_a = -q \int_{\mathbf{r}_a}^{\mathbf{r}_b} \mathbf{E} \cdot d\mathbf{r}$$

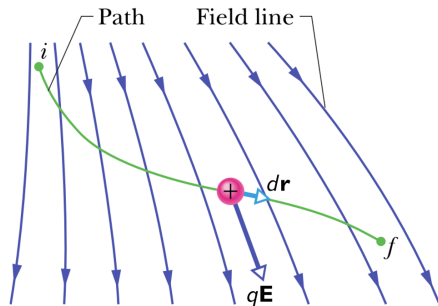


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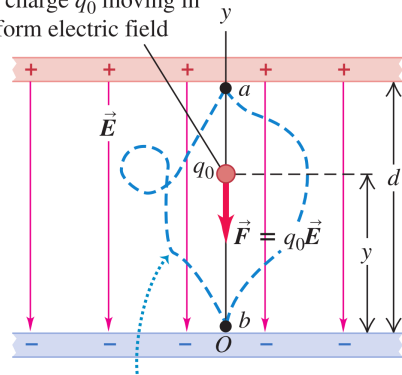
- If, furthermore, $\mathbf{E} = \text{constant}$ along the path taken, then

$$U_b - U_a = -q\mathbf{E} \cdot \int_{\mathbf{r}_a}^{\mathbf{r}_b} d\mathbf{r} = -q\mathbf{E} \cdot [\mathbf{r}]_{\mathbf{r}_a}^{\mathbf{r}_b} = -q\mathbf{E} \cdot (\mathbf{r}_b - \mathbf{r}_a)$$

or

$$U_b - U_a = -q\mathbf{E} \cdot \mathbf{r}_{ab}$$

Point charge q_0 moving in a uniform electric field



The work done by the electric force is the same for any path from a to b :

$$W_{a \rightarrow b} = -\Delta U = q_0 E d$$

Electric Potential Energy

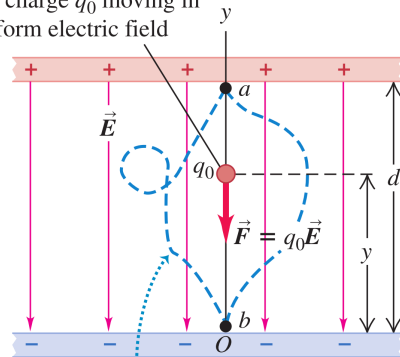
For the specific case of the figure we have

$$\begin{aligned}U_b - U_a &= -q\mathbf{E} \cdot \mathbf{r}_{ab} \\&= -q \left[E(-\hat{\mathbf{j}}) \right] \cdot \left[\underbrace{0\hat{\mathbf{j}}}_b - \underbrace{d\hat{\mathbf{j}}}_a \right] \\&= -qEd \underbrace{\hat{\mathbf{j}} \cdot \hat{\mathbf{j}}}_1 \\&= -qEd\end{aligned}$$

so U decreases, i.e., $U_b < U_a$, because

$$U_b = U_a - qEd$$

Point charge q_0 moving in
a uniform electric field



Important Point!

- In any computation which involves an integral over a curve with the infinitesimal displacement $d\mathbf{r}$, regardless of the direction of the curve, DO NOT CHANGE THE DIRECTION (SIGN) OF $d\mathbf{r}$.
- **Change of integral limits** due to change of direction of movement on the curve **will take care of the sign!**